

## SET 19

1. R: Create 15 ages and divide them into three equal-frequency groups using quantile-based cut points.

The screenshot shows the RStudio interface with the following code in the console:

```
> # Create 15 ages
> ages <- c(18, 21, 23, 25, 27, 29, 31, 34, 36, 39,
+ 42, 45, 48, 52, 60)
>
> # Calculate quantile cut points
> cut_points <- quantile(ages, probs = c(0, 1/3, 2/3, 1))
>
> # Divide ages into 3 equal-frequency groups
> groups <- cut(ages,
+ breaks = cut_points,
+ include.lowest = TRUE,
+ labels = c("Group 1", "Group 2", "Group 3"))
>
> # Display results
> print(cut_points)
0% 33.33333% 66.66667% 100%
18.00000 28.33333 40.00000 60.00000
> print(data.frame(Age = ages, Group = groups))
  Age Group
1  18 Group 1
2  21 Group 1
3  23 Group 1
4  25 Group 1
5  27 Group 1
6  29 Group 2
7  31 Group 2
8  34 Group 2
9  36 Group 2
10 39 Group 2
11 42 Group 3
12 45 Group 3
13 48 Group 3
14 52 Group 3
15 60 Group 3
```

The Environment pane on the right shows the following values:

| Variable   | Value                                                         |
|------------|---------------------------------------------------------------|
| ages       | num [1:15]: 18 21 23 25 27 29 31 34 36 39 ...                 |
| cut_points | Named num [1:4]: 18 28.3 40 60                                |
| groups     | Factor w/ 3 levels "Group 1","Group 2",...: 1 1 1 1 1 2 2 ... |

2. R: Perform equal-width discretization of 10,14,18,23,29,34,41,48,55,63 into four intervals and show frequencies.

The screenshot shows the RStudio interface with the following code in the console:

```
> # Given data
> x <- c(10, 14, 18, 23, 29, 34, 41, 48, 55, 63)
>
> # Find minimum and maximum
> min_x <- min(x)
> max_x <- max(x)
>
> # Calculate equal-width interval width
> width <- (max_x - min_x) / 4
>
> # Create 4 equal-width intervals
> breaks <- seq(min_x, max_x, length.out = 5)
>
> # Discretize the data
> groups <- cut(x, breaks = breaks, include.lowest = TRUE)
>
> # Display intervals and frequencies
> print(groups)
[1] [10,23.2] [10,23.2] [10,23.2] [10,23.2] (23.2,36.5] (23.2,36.5] (36.5,49.8]
[8] (36.5,49.8] (49.8,63] (49.8,63]
Levels: [10,23.2] (23.2,36.5] (36.5,49.8] (49.8,63]
> print(table(groups))
groups
[10,23.2] (23.2,36.5] (36.5,49.8] (49.8,63]
      4          2          2          2
```

The Environment pane on the right shows the following values:

| Variable | Value                                                         |
|----------|---------------------------------------------------------------|
| breaks   | num [1:5]: 10 23.2 36.5 49.8 63                               |
| groups   | Factor w/ 4 levels "[10,23.2)","(23.2,36.5)",...: 1 1 1 1 ... |
| max_x    | 63                                                            |
| min_x    | 10                                                            |
| width    | 13.25                                                         |
| x        | num [1:10]: 10 14 18 23 29 34 41 48 55 63                     |

### 3. WEKA: Create retail ARFF data and use Discretize to convert numeric Income into five bins; apply Apriori.

#### Dataset: Retail Customer Discretization

```
@relation retail_customer
@attribute Age numeric
@attribute Income numeric
@attribute Visits numeric
@attribute Purchase {Yes,No}

@data
21,22000,4,No
25,28000,6,No
29,35000,8,Yes
33,42000,10,Yes
38,50000,12,Yes
45,62000,15,Yes
52,75000,16,Yes
27,30000,7,No
```

The screenshot shows the Weka Explorer application window. The 'Preprocess' tab is selected. The 'Filter' dropdown is set to 'Discretize -B 5 -M -1.0 -R 2 -precision 6'. The 'Current relation' is 'retail\_customer-weka.filters.unsupervised.attribute.Discret...'. The 'Attributes' list on the left shows 'Purchase' selected. The 'Selected attribute' panel on the right displays the following information:

| No. | Label | Count | Weight |
|-----|-------|-------|--------|
| 1   | Yes   | 5     | 5      |
| 2   | No    | 3     | 3      |

Below the table, the 'Class: Purchase (Nom)' is shown. A bar chart visualizes the data with a blue bar for 'Yes' (count 5) and a red bar for 'No' (count 3). The status bar at the bottom indicates 'OK'.

### 4. WEKA: Apply FP-Growth to the discretized retail data and compare rule count/confidence with Apriori.

#### Dataset: Retail Customer – FP-Growth

Use the discretized Retail Customer dataset from Set 19 Q3 after applying WEKA's Discretize filter.

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

FilterChooseDiscretize -B 5 -M -1.0 -R 2 -precision 6ApplyStop

Current relation

Relation: retail\_customer-weka.filters.unsupervised.attribute.Discret...  
Instances: 8

Attributes: 4  
Sum of weights: 8

Attributes

AllNoneInvertPattern

| No. | Name                              |
|-----|-----------------------------------|
| 1   | <input type="checkbox"/> Age      |
| 2   | <input type="checkbox"/> Income   |
| 3   | <input type="checkbox"/> Visits   |
| 4   | <input type="checkbox"/> Purchase |

Remove

Selected attribute

Name: Age  
Missing: 0 (0%)

Distinct: 8

Type: Numeric  
Unique: 8 (100%)

| Statistic | Value  |
|-----------|--------|
| Minimum   | 21     |
| Maximum   | 52     |
| Mean      | 33.75  |
| StdDev    | 10.593 |

Class: Purchase (Nom)Visualize All

5

3

2136.552

Status  
OK

Log x 0